NOTE: THE FOLLOWING PROGRAMS ARE ONLY FOR REFERENCE BASED.
CHECK THE DEVICES WITH FOLLOWING PROGRAMS AFTER COMPLETED THEORY CLASS.

1). LED BLINKING WITH ALL PORT PINS(BY USING THIS PROGRAM USER CAN CHECK PORT PINS ARE WORKING OR NOT)

```
#include<lpc214x.h>

typedef unsigned int u32;

void delay_ms(u32 dlyMS)
{
        dlyMS*=12000;
        while(dlyMS--);
}

int main( )
{
                IODIR0=0XFFFFFFFF;
                IODIR1=0XFFFF0000;
        {
                while(1)
                {
                IOSET0=0XFFFFFFFF;
                IOSET1=0XFFFF0000;
                delay_ms(100);
                IOCLR0=0XFFFFFFFF;
                IOCLR1=0XFFFF0000;
                delay_ms(100);
                }
    }
}
```

2) ACTIVE LOW SWITCH AND ACTIVE LOW LED

```
#include<lpc214x.h>

#define SW     (1<<16)
#define LED   (1<<17)

int main( )
{
        IODIR1=LED;

        while(1)
        {
                if(IOPIN1 & SW)

                    IOSET1=LED;
                else
                    IOCLR1=LED;
        }
}
```

3) ACTIVE HIGH SWITCH AND ACTIVE HIGH LED

```
#include<lpc21xx.h>

#define SW      1<<4
#define LED    1<<5
int main( )
{
        IODIR0=LED;
        while(1)
        {
                if(IOPIN0 & SW)

                    IOSET0=LED;
                 else
                    IOCLR0=LED;

        }
}
```

4. DISPLAY 0-9 ON SINGLE SEGMENT

```
#include<lpc21xx.h>

typedef unsigned int u32;

void delay_ms(unsigned int);

unsigned int i,j,k;
unsigned int a[10]={0xc0,0xf9,0xa4,0xb0,0x99,0x92,0x82,0xf8,0x80,0x90};

int main( )
{
        IODIR0=0X000000FF;
        while(1)
        {
                for(i=0;i<10;i++)
                  {
                    IOPIN0=a[i];
                    delay_ms(1000);
                  }
        }
}


void delay_ms(u32 dlyMS)
{
        dlyMS*=12000;
        while(dlyMS--);
}
```

5. DISPLAY 45 ON TWO SEGMENTS WITH MULTIPLEXING MODE

```
#include<lpc214x.h>

typedef unsigned char u8;
typedef unsigned int u32;

void delay_ms(unsigned int);

#define SETBIT(WORD,BITPOS)  (WORD|=(1<<BITPOS))
#define CLRBIT(WORD,BITPOS)  (WORD&=~(1<<BITPOS))
#define WRITEBYTE(WORD,BITPOS,BYTE) (WORD=(WORD&~(255<<BITPOS))|((BYTE)<<BITPOS))

#define CA7SEGMUX 0

#define SEL1 8
#define SEL2 9

const u8 seg_lut[10]={0xc0,0xf9,0xa4,0xb0,0x99,0x92,0x82,0xf8,0x80,0x90};
//unsigned char seg_lut[]={0xc0,0xf9,0xa4,0xb0,0x99,0x92,0x82,0xf8,0x80,0x90};

int main()
{
        u8 n=45;

        WRITEBYTE(IODIR0,CA7SEGMUX,0XFF);
        SETBIT(IODIR0,SEL1);
        SETBIT(IODIR0,SEL2);

        while(1)
        {
                SETBIT(IOSET0,SEL1);
                WRITEBYTE(IOPIN0,CA7SEGMUX,seg_lut[n/10]);
                delay_ms(1);
                SETBIT(IOCLR0,SEL1);

                SETBIT(IOSET0,SEL2);
                WRITEBYTE(IOPIN0,CA7SEGMUX,seg_lut[n%10]);
                delay_ms(1);
                SETBIT(IOCLR0,SEL2);
        }

}
void delay_ms(u32 dlyMS)
{
    dlyMS*=12000;
    while(dlyMS--);
}
```